FarmerML

Farmer Income Prediction & Credit Risk Modelling

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1. EDA Insights

Target Variable — Income Distribution

- Raw income is heavily right-skewed (skewness printed); log(1+x) transform normalises it — visualised as raw histogram, log histogram, and boxplot.
- Median, mean, and 90th percentile computed to understand income spread across farmers.

Key Numerical Feature Distributions

- Histograms with annotated skewness for: Total Land, K022/R022 Rainfall, Mandi Proximity, Non-Agri Income.
- Land has a long right tail → log-transform needed. Mandi proximity is bimodal → inverse market-access-score justified.

Categorical Breakdowns vs Income

- Median income compared across REGION, SEX, MARITAL_STATUS via bar charts.
- Village Agri Category (Poor/Average/Good) shows a clear monotone income gradient → ordinal encoding (0/1/2) over one-hot confirmed.

Correlation Matrix & Scatter Plots

- Top 15 numerical predictors correlated with target — lower-triangle heatmap (coolwarm, annot=True).
- Top 4 correlates scatter-plotted vs log(income) with Pearson r. log(Land) vs log(Income) is more linear → log1p(land) feature justified.

Seasonal Rainfall Patterns

- Boxplots and mean-bar charts for all 5 seasons (K022, R022, K021, R021, R020).
- Pearson r(K022 rain, log income) computed and printed. High cross-farmer variance → volatility & deviation features needed.

EDA Observation	Modelling Implication
Income heavily right-skewed	Predict log(1+income); exponentiate back
Good villages earn significantly more	Ordinal encode village category (0/1/2)
log(Land)–log(Income) more linear	Use log1p(land) as feature
Mandi proximity bimodal	market_access_score = 1 / (1 + km)
High Kharif rainfall variance	Rainfall volatility & deviation features
Columns >70% missing found	Drop before imputation pipeline

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2. Data Preparation & Feature Engineering

2.1 Missing Value Strategy

- **Drop:** Columns with >70% missing dropped entirely.
- **KNN Imputer (k=5):** Features with 10–30% missingness — preserves local structure over global stats.
- **Mean/Median:** Residual numerics — mean if |skew|<1, else median. Categoricals filled by mode.
- **Second pass:** Identical imputation re-run after feature engineering to handle newly introduced NaNs.

2.2 Outlier Treatment

All numerical columns capped at 3×IQR fence (Q1–3×IQR, Q3+3×IQR). Applied twice — pre and post feature engineering — to suppress data-entry errors while preserving genuine high earners within the fence.

2.3 Encoding Choices

Feature	Method	Reason
Village Agri Category	Ordinal (Poor=0, Avg=1, Good=2)	Monotone income gradient confirmed in EDA
SEX / MARITAL_STATUS	Binary flags: is_male, is_married	Binary variables; avoids extra dummy columns
REGION	One-hot with prefix "region_"	Unordered; low cardinality
Low-cardinality cats (≤8 unique)	One-hot, drop_first=False	Preserves all category signals
State, DISTRICT, VILLAGE	Dropped	Too high cardinality; leakage/noise risk

2.4 Engineered Features

Feature	Formula	Rationale
land_log	log1p(Total_Land)	Normalises skew; log-log linear with income
market_access_score	1 / (1 + mandi_km)	Bounded [0,1]; proximity benefit
land_market_interact	land_log × market_access_score	Productive land near markets amplified
rainfall_mean / volatility / cv	mean, std, std/mean across 5 seasons	Baseline climate + instability proxy
rainfall_dev_k22 / r22	season_rain – rainfall_mean	Deviation from farmer's personal average
kharif / rabi _yoy_delta	K022–K021 / R022–R021	Year-on-year growing season shift
drought_season_count	Seasons below 25th pct rainfall	Cumulative drought exposure per farmer
kharif_rabi_ratio	mean(K) / (mean(R) + 1e-6)	Seasonal crop balance indicator
income_volatility_score	rain_std / (rain_mean + 1e-6)	Credit risk proxy — used in Bonus B
trange per season	Tmax – Tmin (parsed from string)	Thermal stress per growing season
log_non_agri	log1p(Non_Agri_Income)	Income diversification signal
land_per_income_proxy	land / (Non_Agri + 1)	Land utilisation vs off-farm income

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3. Model Selection Rationale & Results

3.1 Baseline — Linear Regression (StandardScaler Pipeline)

Scikit-learn Pipeline (StandardScaler → LinearRegression) trained on log(1+income). Establishes a performance floor. Train and 20% hold-out MAPE both computed and printed.

3.2 Advanced — XGBoost + Optuna (30 trials, 5-fold CV)

XGBoost selected because: (1) captures non-linear interactions (land × market access × rainfall) that linear models miss, (2) robust to post-capping outliers, (3) handles sparse one-hot features natively. Optuna minimised 5-fold CV MAPE over 30 trials.

Hyperparameter	Search Range	Hyperparameter	Search Range
n_estimators	300 – 800	min_child_weight	1 – 6
learning_rate	0.01 – 0.08 (log)	reg_alpha	0.0 – 3.0
max_depth	4 – 8	reg_lambda	1.0 – 8.0
subsample	0.7 – 1.0	colsample_bytree	0.6 – 1.0

3.3 Performance Summary

Model	Train MAPE	Val/Test MAPE	Gap	Verdict
Linear Regression (Baseline)	0.2742	0.2698	-0.0043	Fine
XGBoost (Tuned)	0.1402	0.2049	0.0648	Better accuracy

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4. Bonus Analyses

Bonus A — Intervention Recommender (Bottom-25% Farmers)

- SHAP TreeExplainer applied on full training set. Bottom 25% farmers identified by predicted income (threshold printed).
- Fixed variables excluded from recommendations: rainfall_mean/volatility/cv, deviation cols, drought_season_count, kharif_rabi_ratio.
- Remaining features ranked by mean |SHAP| within the bottom-25% — top 2 are the actionable levers.
- Counterfactual uplift: each lever shifted to its 75th percentile; average % income gain computed and printed.

Bonus B — Seasonal Volatility Modelling

- K022/R022/K021/R021/R020 reshaped into a farmer × season × year panel (not flattened). YoY delta computed per farmer per season.
- Per-farmer: rain_mean, rain_std, rain_cv (coefficient of variation), income_quartile. Scatter + bar charts: CV vs log(income); median CV by quartile Q1–Q4.
- Kharif/Rabi split volatility: |kharif_share – 0.5| computed — balanced farmers shown to have more stable incomes.

- Drought frequency: fraction of seasons below P30 rainfall correlated with log(income) — r and p-value printed.
- **Conclusion:** volatile and drought-prone farmers are a riskier credit bet regardless of average income.

Final Model: XGBoost (Optuna, 30 trials, 5-fold CV) | **Metric:** MAPE | **Target:** log(1+income), exponentiated back | **Output:** Colin_predictions.csv